Topology and Geometry in Physics – Note 2

Ling-Wei Luo*

School of Fundamental Physics and Mathematical Sciences,
Hangzhou Institute for Advanced Study, UCAS

June 19, 2026

Abstract

This is the *second* note on the course of topology and geometry in physics at the School of Fundamental Physics and Mathematical Sciences, Hangzhou Institute for Advanced Study, UCAS, China. In this note we will learn the differential forms in the first part from Sec. 1. Particularly, I will also show you how to express the electromagnetism by differential forms in theoretical physics. The second part is the Sec. 2, we will learn the method of Cartan's exterior covariant differetiation and moving frame method.

Contents

*Electronic address: lwluo.phys@gmail.com

1 Differentiable Manifolds

1.1 Manifolds

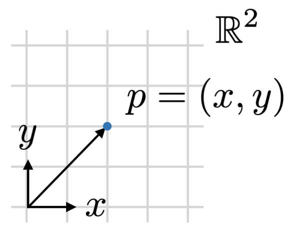


Fig. 1.1: A point p in a Euclidean space $\mathbb{R}^2$.

Euclidean space. We start the discussion on an Euclidean space $\mathbb{R}^2$. Given a point $p \in \mathbb{R}^2$, we can parameterize a point by two parameters x and y as coordinates, i.e., a point p is always written by $p = (x, y) = (x^1, x^2)$. For $\mathbb{R}^n$, a point p can be represented by the coordinates $(x^1, x^2, \cdots, x^n)$.

Hausdorff space. A topological space is called a *Hausdorff space* if for any two distinct points p and q, there exists two neighborhoods U and V for $p \in U$ and $q \in V$ such that the intersection of U and V is empty, i.e., $U \cap V = \varnothing$.

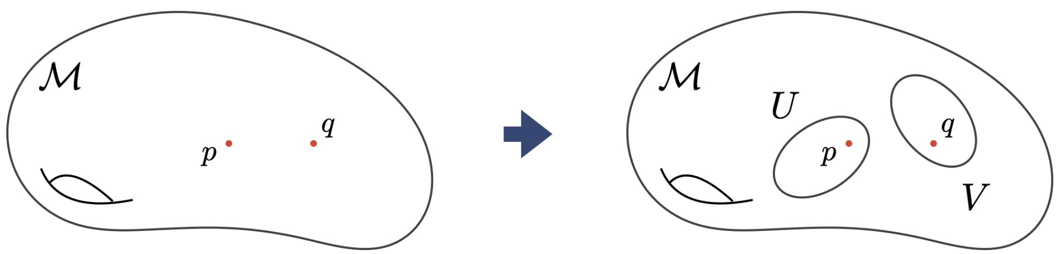


Fig. 1.2: A Hausdorff space $\mathcal{M}$.

Manifold. Let $\mathcal{M}$ be a Hausdorff space. We say that $\mathcal{M}$ is called an n-dimensional *topological manifold* if the neighborhood of any point of $\mathcal{M}$ is isomorphic to an open set in Euclidean space $\mathbb{R}^n$.

Morphism. We will call a map the *morphism*, if it is *structure preserving*.

Local coordinates. Consider a point p in a neighborhood U of a manifold $\mathcal{M}$ with dimension n, we can define a local coordinate system (or a *coordinate chart* or a *coordinate patch* or a *parametrization*) of a neighborhood U of a point p and a *homeomorphism* (an *one-to-one* and *onto* map) φ as a pair (U, φ). The local coordinates of p is given by a map φ as a set of n-tuple real numbers in a neighborhood $D \subset \mathbb{R}^n$ as

$$\varphi : U \longrightarrow \mathbb{R}^n \tag{1.1a}$$

$$p \longmapsto \varphi(p) \coloneqq x = (x^1, x^2, \cdots, x^n) \in D. \tag{1.1b}$$

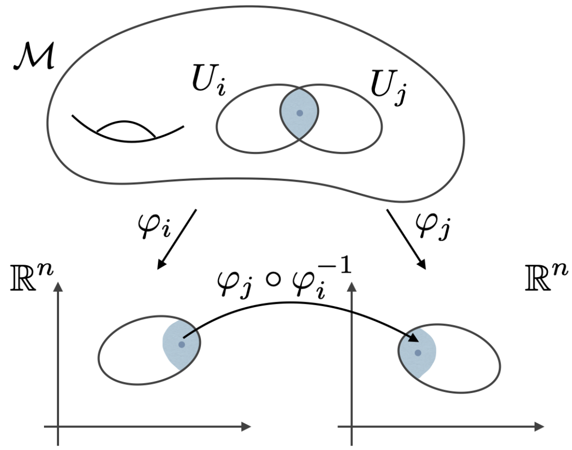


Fig. 1.3: A manifold with coordinate charts.

Coordinate functions. The projection map

$$u^\mu : \mathbb{R}^n \longrightarrow \mathbb{R} \tag{1.2}$$

sending the coordinate $x = (x^1, x^2, \cdots, x^n)$ of a point p to its μth-component is called the *coordinate function*. Therefore, we can construct a composition

$$\varphi^\mu := u^\mu \circ \varphi : U \longrightarrow \mathbb{R} \tag{1.3}$$

such that the coordinate $x^\mu = u^\mu(x) = u^\mu \circ \varphi(p) \equiv \varphi^\mu(p)$.

Coordinate transformation. Consider two local coordinate systems (U_i, φ_i) and (U_j, φ_j) on $\mathcal{M}$. A point p in $U_i \cap U_j$ has two local coordinates $\varphi_i(p) = x_i$ and $\varphi_j(p) = x_j$. We define the *transition function* between ith neighourhood and jth neighourhood

$$\begin{cases} \varphi_{ji} := \varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \longrightarrow \varphi_j(U_i \cap U_j), & \text{(1.4a)} \\ \varphi_{ij} := \varphi_i \circ \varphi_j^{-1} : \varphi_j(U_i \cap U_j) \longrightarrow \varphi_i(U_i \cap U_j), & \text{(1.4b)} \end{cases}$$

and

$$\begin{cases} x_i \longmapsto x_j = \varphi_j \circ \varphi_i^{-1}(x_i) = \varphi_{ji}(x_i), & \text{(1.5a)} \\ x_j \longmapsto x_i = \varphi_i \circ \varphi_j^{-1}(x_j) = \varphi_{ij}(x_j). & \text{(1.5b)} \end{cases}$$

More precisely, the coordinates are related by

$$\begin{cases} x^\mu \longmapsto x^\nu = u_j^\nu(x_j) = u_j^\nu \circ \varphi_j \circ \varphi_i^{-1}(x_i) = u_j^\nu \circ \varphi_{ji}(x_i), & \text{(1.6a)} \\ x^\nu \longmapsto x^\mu = u_i^\mu(x_i) = u_i^\mu \circ \varphi_i \circ \varphi_j^{-1}(x_j) = u_i^\mu \circ \varphi_{ij}(x_j). & \text{(1.6b)} \end{cases}$$

Example. (A circle S^1 in $\mathbb{R}^2$).

Define a n-dimensional unit sphere by

$$S^n = \{x \in \mathbb{R}^{n+1} \mid (x^1)^2 + (x^1)^2 + \cdots + (x^{n+1})^2 = 1\}. \tag{1.7}$$

So, $n = 1$ gives a circle with the radius $r = 1$ in $\mathbb{R}^2$.

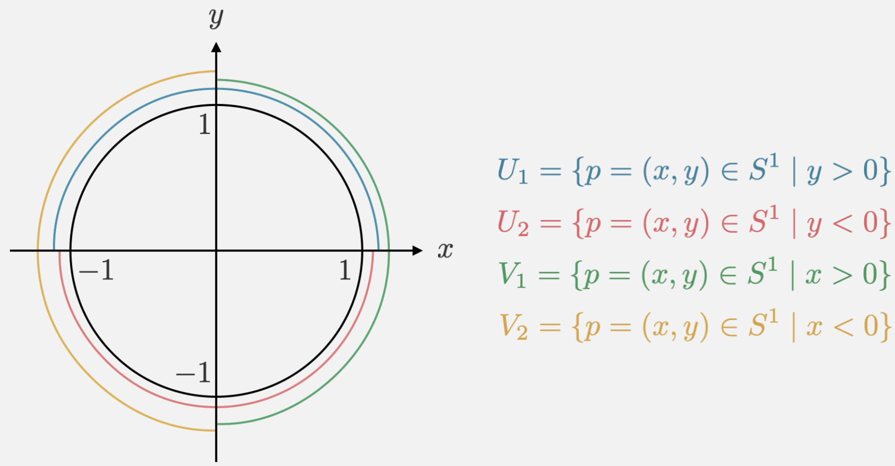


Fig. 1.4: A circle with four charts.

Let homeomorphisms be

$$\begin{cases} \varphi_{U_1}(p) = x^1, \\ \varphi_{U_2}(p) = x^1, \\ \varphi_{V_1}(p) = x^2, \\ \varphi_{V_2}(p) = x^2. \end{cases} \tag{1.8}$$

It is obvious that we have $U_1 \cup U_2 \cup V_1 \cup V_1 = S^1$. In the intersection $U_1 \cap V_2$, we have

$$\begin{cases} x^2 = \sqrt{1 - {x^1}^2} > 0, \\ x^1 = -\sqrt{1 - {x^2}^2} < 0. \end{cases} \tag{1.9}$$

It is obvious that these two functions are both smooth (C^∞) functions, thus (U_1, φ_{U_1}) and (V_2, φ_{V_2}) are C^∞-compatible.

Differentiable manifold. The collection of all local coordinate systems is called an *atlas* of a manifold $\mathcal{M}$ defined by $\mathcal{A} = \{(U_i, \varphi_i)\}$. For any intersection of $U_i \cap U_j$, if φ_{ij} is continuously differentiable r times, then the atlas $\mathcal{A}$ is called the coordinate neighborhood system of class C^r and any pair of charts in $\mathcal{A}$ are called C^r-compatible. Therefore, if all the transformation functions are smooth (or infinitely differentiable), we say that the manifold is smooth and the atlas $\mathcal{A}$ with the differentiability defines the *differential structure of class* C^∞ on the topological manifold $\mathcal{M}$.

Diffeomorphism. Consider another manifold $\mathcal{N}$, the diffeomorphism is a homeomorphism $f : \mathcal{M} \longrightarrow \mathcal{N}$ such that f and f^{-1} are both smooth. This is *globally* defined. *Locally*, given two local coordinate systems (U_i, φ_i) and (V_j, ψ_j) on $\mathcal{M}$ and $\mathcal{N}$, respectively, the induced coordinate

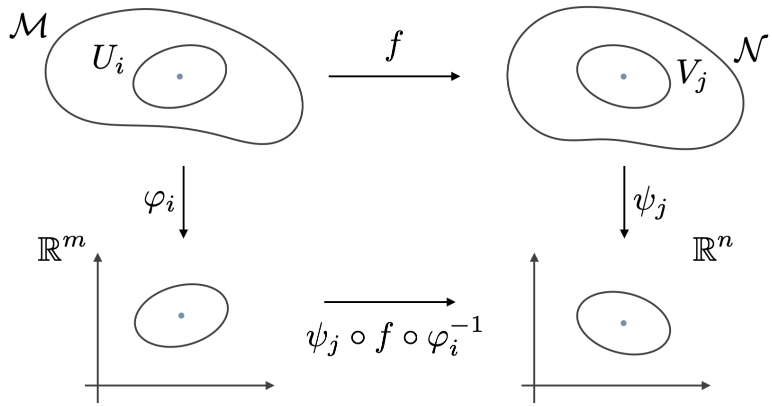


Fig. 1.5: A map from manifold $\mathcal{M}$ to $\mathcal{N}$ with coordinate charts.

transformation of the *local* diffeomorphism is read as

$$\begin{cases} x_i \longmapsto y_j = \psi_j \circ f \circ \varphi_i^{-1}(x_i) \coloneqq f_{ji}^{(yx)}(x_i), & (1.10a) \\ y_j \longmapsto x_i = \varphi_i \circ f^{-1} \circ \psi_j^{-1}(y_j) \coloneqq f_{ij}^{(xy)}(y_j). & (1.10b) \end{cases}$$

We note that $f_{ji}^{(yx)} = \psi_j \circ f \circ \varphi_i^{-1}$ and $f_{ij}^{(xy)} = \varphi_i \circ f^{-1} \circ \psi_j^{-1}$ are called the *expressions* of f in the local coordinates φ_i and ψ_j.

1.2 Tangent Vector and Tangent Vector Space

Vector fields in $\mathbb{R}^n$ Let f be a function over $\mathbb{R}$

Define a curve $\gamma : I_\varepsilon = (-\varepsilon, \varepsilon) \subset \mathbb{R} \longrightarrow \mathcal{M}$, which gives $\tau = \gamma(t)$ on a smooth manifold $\mathcal{M}$. Locally a point is given by $p = \gamma(0)$ on U and the corresponding parameterized coordinates should be

$$\varphi(\gamma(t)) = \varphi \circ \gamma(t) = x(t). \tag{1.11}$$

We assume that $f \in \mathfrak{F}(\mathcal{M}) : \mathcal{M} \longrightarrow \mathbb{R}$ is a function on $\mathcal{M}$ where $\mathfrak{F}(\mathcal{M})$ is the algebra of C^k. So the coordinate expression of f on U can be given by

$$f(p) = f\big(\varphi^{-1}(x)\big) = f \circ \varphi^{-1}(x) \coloneqq F(x). \tag{1.12}$$

The maps including the coordinate chart from $\mathbb{R}$ to $\mathcal{M}$ to $\mathbb{R}$ is shown in Fig. 1.6. Combine (1.11) and (1.12), a one-parameter varying function is

$$f\big(\gamma(t)\big) = f \circ \gamma(t) = \underbrace{f \circ \varphi^{-1}}_{F} \circ \underbrace{\varphi \circ \gamma}_{x}(t) = F \circ x(t) = F\big(x(t)\big). \tag{1.13}$$

A *tangent* vector or *velocity* vector filed at point $\gamma(t)$ with the local coordinates $\{x^\mu(t)\}$ in U can be obtained by the so-called derivative map

$$d\gamma : T_t\mathbb{R} \longrightarrow T_{\gamma(t)}\mathcal{M} \tag{1.14a}$$

$$\frac{d}{dt} \longmapsto d\gamma\left(\frac{d}{dt}\right) \equiv \frac{d\gamma}{dt} = \dot{\gamma}(t) \coloneqq X_{\gamma(t)}. \tag{1.14b}$$

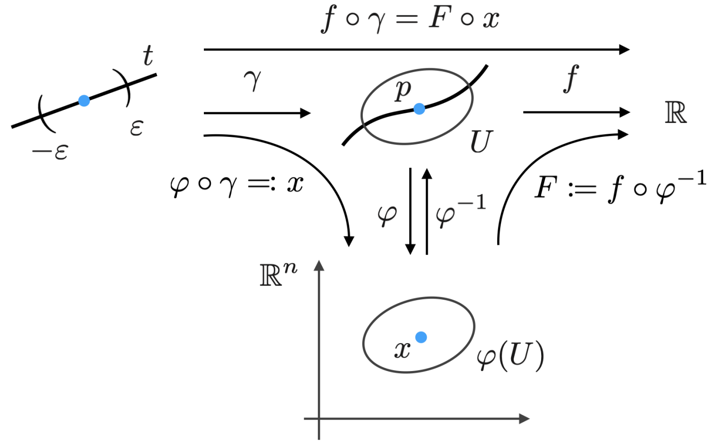


Fig. 1.6: A map of a function f from $\mathbb{R}$ to $\mathbb{R}$.

The tangent vector act on f would be

$$\frac{df\big(\gamma(0)\big)}{dt} = \frac{dF\big(x(0)\big)}{dt} = \frac{dx^\mu}{dt}(0) \cdot \frac{\partial F}{\partial x^\mu}\bigg|_{\gamma(0)}, \tag{1.15}$$

where $\partial/\partial x^\mu \equiv \partial_\mu$. Hence, a vector field X at $p = \gamma(0)$ is defined as a map which sends a function $f \in \mathfrak{F}(\mathcal{M})$ to $\mathbb{R}$

$$X_p : \mathfrak{F}(\mathcal{M}) \longrightarrow \mathbb{R}, \tag{1.16}$$

and it has an expression of

$$X_p = X_p^\mu \, (\partial_\mu)_p, \tag{1.17}$$

where the coefficients can be obtained by applying X to the coordinates x^μ, i.e.,

$$\boxed{X_p^\mu = X_p(x^\mu)} = X_p^\nu \frac{\partial x^\mu}{\partial x^\nu} = X_p^\nu \delta_\nu^\mu. \tag{1.18}$$

It can obviously find that $X_p := X_{\gamma(0)}$ is a *derivation* of a smooth manifold $\mathcal{M}$ at point $\gamma(0) = p$, the set of all derivations is denoted by $D_p^r\mathcal{M}$ and

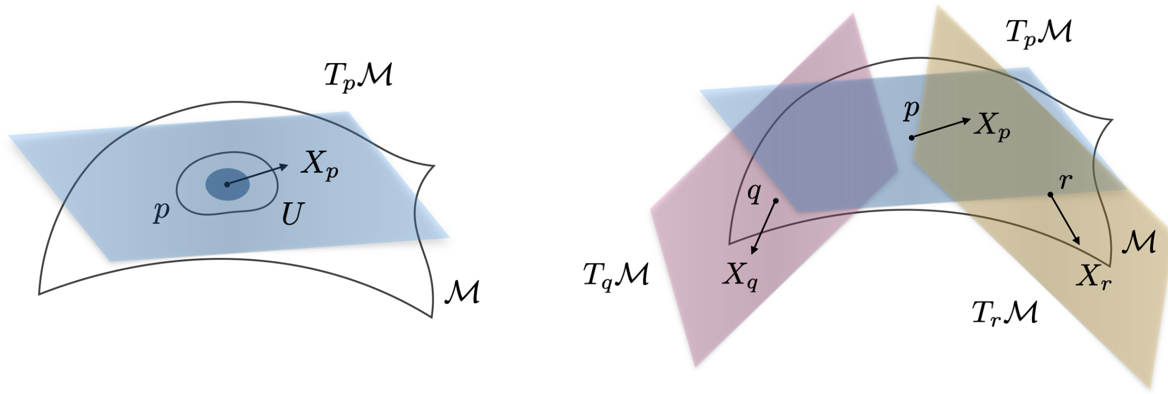
$$X_{\gamma(t)}(f) = X^\mu(t)\, \partial_\mu F|_{\gamma(t)}. \tag{1.19}$$

Therefore, by identification of $T_{\gamma(t)}\mathcal{M}$ and $D_{\gamma(t)}\mathcal{M}$, components of a tangent vector are given by a set of differential equations

$$X^\mu(t) := \frac{dx^\mu}{dt}(t) \tag{1.20}$$

So the solution x^μ of (1.20) would be obtained

$$\boxed{x^\mu(t) = x^\mu(0) + \int_0^t dt'\, X^\mu(t')} = x^\mu(0) + t\, X_p^\mu + \mathcal{O}(t^2), \tag{1.21}$$



(a) A tangent space $T_p\mathcal{M}$ over a point p.

(b) Tangent spaces $T_p\mathcal{M}$, $T_q\mathcal{M}$ and $T_r\mathcal{M}$ over three point p, q and r respectively.

Fig. 1.7: Tangent space.

where $X^\mu(t') = X^\mu(0) + t'\, dX^\mu(t')/dt' + \mathcal{O}(t'^2)$ and $X_p^\mu := X^\mu(0)$. Therefore, the tangent vector of the curve $\gamma(t)$ at p is a directional derivative along the direction of X

$$\dot{\gamma}(0) \equiv X_p = X_p^\mu(\partial_\mu)_p = X^\mu(0)(\partial_\mu)_{\gamma(0)}. \tag{1.22}$$

The discussion above proves that every vector is a linear combination of $\{\partial_\mu\}$. This implies that a vector X represents a differential operator on $\mathcal{M}$. The basis vectors $\{\partial_\mu\}$ given by the local coordinates is called a *natural* basis. Moreover, we can define a so-called *tangent bundle* which is a collection of all tangent spaces

$$T\mathcal{M} = \bigcup_{p\in\mathcal{M}} T_p\mathcal{M}, \tag{1.23}$$

and the set of all *differential* (or *smooth*) vector fields on $\mathcal{M}$ is always denoted by $\mathfrak{X}(\mathcal{M}) = \Gamma(T\mathcal{M})$.

Consider the bracket operation of vectors X and $Y \in \mathfrak{X}(\mathcal{M}) = \Gamma(T\mathcal{M})$ acting on $f \in \mathfrak{F}(\mathcal{M})$, then we have

$$\begin{aligned} [X,Y]f &= [X^\mu\partial_\mu, Y^\nu\partial_\nu]f \\ &= X^\mu(\partial_\mu Y^\nu)(\partial_\nu f) - Y^\nu(\partial_\nu X^\mu)(\partial_\mu f) \\ &= \underbrace{[X^\mu(\partial_\mu Y^\nu) - Y^\mu(\partial_\mu X^\nu)]}_{[X,Y]^\nu}(\partial_\nu f), \end{aligned} \tag{1.24}$$

which shows that $[X,Y]$ is a vector field. It implies that $\mathfrak{X}(\mathcal{M})$ is a Lie algebra. In addition we have the *Jacobi's identity*

$$[[X,Y],Z] + [[Y,Z],X] + [[Z,X],Y] = 0. \tag{1.25}$$

1.3 Dual Vector Space and Cotangent Space

Let f be a function over a field $\mathfrak{F}$. A field $\mathfrak{F}$ can be either $\mathbb{R}$ or $\mathbb{C}$. We call f is *linear* if

$$f(ax+by) = af(x) + bf(y), \quad a,b \in \mathfrak{F}. \tag{1.26}$$

Let V be a n-dimensional vector space and a map $\vec{f}^* : V \longrightarrow \mathfrak{F}$ is a $\mathfrak{F}$-valued *function* on V. For $\vec{v}_1, \vec{v}_2 \in V$ and $a_1, a_2 \in \mathfrak{F}$ such that

$$\vec{f}^*(a_1\vec{v}_1 + a_2\vec{v}_2) = a_1\vec{f}^*(\vec{v}_1) + a_2\vec{f}^*(\vec{v}_2), \tag{1.27}$$

we call $\vec{f}^*$ is a $\mathfrak{F}$-valued linear function on V. The collection of all $\vec{f}^*$s on V forms a n-dimensional vector space, which is called *dual vector space* of V, denoted by V^*. Let $\{\vec{e}_i\}$ be a set of of basis vector of V, a vector can be written by

$$\vec{v} = \vec{e}_i\, v_i. \tag{1.28}$$

Let $\vec{f}^* \in V^*$ such that

$$\vec{f}^*(v) = \vec{f}^*(\vec{e}_i\, v_i) = \vec{f}^*(\vec{e}_i)v_i. \tag{1.29}$$

Define a set $\{\vec{e}_i^*\}$ to be *linear functions* of V such that

$$\vec{e}_i^*(\vec{e}_j) = \delta_{ij}, \tag{1.30}$$

which can be called the *dual relation*, and therefore

$$\vec{e}_i^*(v) = v_i. \tag{1.31}$$

We can define

$$f_i^* \coloneqq \vec{f}^*(\vec{e}_i) \tag{1.32}$$

Hence, we have

$$\vec{f}^*(v) = f_i^*\, v_i = f_i^*\, \vec{e}_i^*(v) \qquad \Longrightarrow \qquad \boxed{\vec{f}^* = f_i^*\, \vec{e}_i^*.} \tag{1.33}$$

New notation. In order to unify the notation, we use $\{e_i\}$ without an arrow as the basis of V and $\{\vartheta^i\}$ as the dual basis of V^* instead of $\{\vec{e}_i^*\}$, while, for the notation of components, the upper and lower indices are, respectively, used for components of a vector and a dual vector. That means we have the following replacement:

$$\begin{cases} \vec{e}_i \quad \longrightarrow \quad e_i \quad \text{and} \quad v_i \quad \longrightarrow \quad v^i, & (1.34\text{a}) \\ \vec{e}_i^* \quad \longrightarrow \quad \vartheta^i \quad \text{and} \quad w_i^* \quad \longrightarrow \quad w_i. & (1.34\text{b}) \end{cases}$$

Hence, the new notation of a vector $\vec{v}$ and a dual vector $\vec{w}^*$ is written, respectively, by

$$\boxed{\vec{v} = e_i\, v^i \in V} \quad \text{and} \quad \boxed{\vec{w}^* = w_i\, \vartheta^i \in V^*} \tag{1.35}$$

and the dual relation (1.30) becomes

$$\boxed{\vartheta^i(e_j) = \delta^i_j.} \tag{1.36}$$

In physics, we always write vectors and dual vectors by their components only, a vector component v^i carries *upper* index called a *contravariant vector*, while a dual vector component w_i carries *lower* index called a *covariant vector* (or *covector* simply):

$$\text{A } \textit{contravariant} \text{ vector: } \vec{v} \quad \longrightarrow \quad v^i \equiv \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix}; \tag{1.37a}$$

$$\text{A } \textit{covariant} \text{ vector: } \vec{w}^* \quad \longrightarrow \quad w_i \equiv \begin{pmatrix} w_1 & w_2 & \cdots & w_n \end{pmatrix}. \tag{1.37b}$$

Alternatively, the following realization of vectors and forms is given. The upper index can be regarded as the index of a column, while the lower index represents a row index, i.e., vectors and covectors are respectively realized by the product of a row matrix and a column matrix

$$\vec{v} = e_i\, v^i \equiv \begin{pmatrix} e_1 & e_2 & \cdots & e_n \end{pmatrix} \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix} \quad \text{and} \quad \vec{w}^* = w_i\, \vartheta^i \equiv \begin{pmatrix} w_1 & w_2 & \cdots & w_n \end{pmatrix} \begin{pmatrix} \vartheta^1 \\ \vartheta^2 \\ \vdots \\ \vartheta^n \end{pmatrix}. \tag{1.38}$$

By definition, a covector is a function of a vector. One can also take the covector as an operator acting on a vector realized by

$$\begin{aligned} \vec{w}^*(\vec{v}) &= w_i\, \vartheta^i(e_j\, v^j) = w_i\, \vartheta^i(e_j) v^j = w_i\, \delta^i_j\, v^j = w_j v^j \\ &= w_1 v^1 + w_2 v^2 + \cdots + w_n v^n \equiv \begin{pmatrix} w_1 & w_2 & \cdots & w_n \end{pmatrix} \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^n \end{pmatrix}. \end{aligned} \tag{1.39}$$

Connection on a cotangent space.

Consider that the differentiation on the both sides of $\vec{e}^{\,*}_i(\vec{e}_j) = \delta_{ij}$:

$$\begin{aligned} 0 = \boldsymbol{d}\Big(\vec{e}^{\,*}_i(\vec{e}_j)\Big) &= (\boldsymbol{d}\vec{e}^{\,*}_i)\vec{e}_j + \vec{e}^{\,*}_i(\boldsymbol{d}\vec{e}_j) \\ &= \Gamma^{*k}{}_i\Big(\vec{e}^{\,*}_k(\vec{e}_j)\Big) + \Gamma^k{}_j\Big(\vec{e}^{\,*}_i(\vec{e}_k)\Big) = \Gamma^{*j}{}_i + \Gamma^i{}_j. \end{aligned} \tag{1.40}$$

Hence, the connection (form) in the dual space is

$$\boxed{\Gamma^{*j}{}_i = -\Gamma^i{}_j.} \tag{1.41}$$

Furthermore, by transpose T, we have

$$\boldsymbol{d}\vec{e}^{\,*}_i = \vec{e}^{\,*}_k\, \Gamma^{*k}{}_i \quad \xrightarrow{\mathsf{T}} \quad \boldsymbol{d}\vartheta^i = \Gamma^{*i}{}_k\, \vartheta^k \quad \Longrightarrow \quad \boxed{{}^{(\text{old})}\Gamma^{*k}{}_i \quad \xrightarrow{\mathsf{T}} \quad {}^{(\text{new})}\Gamma^{*i}{}_k = \left({}^{(\text{old})}\Gamma^{*k}{}_i\right)^{\mathsf{T}}}, \tag{1.42}$$

or, more precisely, in the matrix form

$$\underbrace{\left(\boldsymbol{d}\vec{e}^{*}_1 \quad \boldsymbol{d}\vec{e}^{*}_2 \quad \cdots \quad \boldsymbol{d}\vec{e}^{*}_i \quad \cdots \quad \boldsymbol{d}\vec{e}^{*}_n \right)}_{(\boldsymbol{d}\mathbf{e}^{*}_i)}$$

$$= \underbrace{\left(\vec{e}^{*}_1 \quad \vec{e}^{*}_2 \quad \cdots \quad \vec{e}^{*}_k \quad \cdots \quad \vec{e}^{*}_n \right)}_{(\mathbf{e}^{*}_k)} \underbrace{\begin{pmatrix} \Gamma^{*1}{}_1 & \Gamma^{*1}{}_2 & \cdots & \Gamma^{*1}{}_i & \cdots & \Gamma^{*1}{}_n \\ \Gamma^{*2}{}_1 & \Gamma^{*2}{}_2 & \cdots & \Gamma^{*2}{}_i & \cdots & \Gamma^{*2}{}_n \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Gamma^{*k}{}_1 & \Gamma^{*k}{}_2 & \cdots & \Gamma^{*k}{}_i & \cdots & \Gamma^{*k}{}_n \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Gamma^{*n}{}_1 & \Gamma^{*n}{}_2 & \cdots & \Gamma^{*n}{}_i & \cdots & \Gamma^{*n}{}_n \end{pmatrix}}_{\left(\boldsymbol{\Gamma}^{*k}{}_i\right)} \tag{1.43a}$$

$$\xrightarrow{\mathrm{T}} \underbrace{\begin{pmatrix} \boldsymbol{d}\vartheta^1 \\ \boldsymbol{d}\vartheta^2 \\ \vdots \\ \boldsymbol{d}\vartheta^i \\ \vdots \\ \boldsymbol{d}\vartheta^n \end{pmatrix}}_{(\boldsymbol{d\theta}^i)} = \underbrace{\begin{pmatrix} \Gamma^{*1}{}_1 & \Gamma^{*1}{}_2 & \cdots & \Gamma^{*1}{}_k & \cdots & \Gamma^{*1}{}_{1n} \\ \Gamma^{*2}{}_1 & \Gamma^{*2}{}_2 & \cdots & \Gamma^{*2}{}_k & \cdots & \Gamma^{*2}{}_n \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Gamma^{*i}{}_1 & \Gamma^{*i}{}_2 & \cdots & \Gamma^{*i}{}_k & \cdots & \Gamma^{*i}{}_n \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Gamma^{*n}{}_1 & \Gamma^{*n}{}_2 & \cdots & \Gamma^{*n}{}_k & \cdots & \Gamma^{*n}{}_n \end{pmatrix}}_{\left(\boldsymbol{\Gamma}^{*i}{}_k\right)} \underbrace{\begin{pmatrix} \vartheta^1 \\ \vartheta^2 \\ \vdots \\ \vartheta^k \\ \vdots \\ \vartheta^n \end{pmatrix}}_{(\boldsymbol{\theta}^k)}. \tag{1.43b}$$

It can be check that the connection form in the new notation in the dual vector space is has the same result in terms of the connection form in the vector space. Applying the differentiation on (1.36) leads to

$$\begin{aligned} 0 &= \boldsymbol{d}\vartheta^i(e_j) = (\boldsymbol{d}\vartheta^i)e_j + \vartheta^i(\boldsymbol{d}e_j) \\ &= \Gamma^{*i}{}_k\,\vartheta^k(e_j) + \Gamma^{ki}{}_{j\vartheta}(e_k) = \Gamma^{*i}{}_j + \Gamma^i{}_j \qquad \Longrightarrow \qquad \boxed{\Gamma^{*i}{}_j = -\Gamma^i{}_j.} \end{aligned} \tag{1.44}$$

More precisely, it is read by

$$\boxed{{}^{(\mathrm{new})}\Gamma^{*i}{}_j = {}^{(\mathrm{ond})}\Gamma^{*j}{}_i = -\Gamma^i{}_j,} \tag{1.45}$$

which is consist with (1.41).

Coordinate 1-forms. The map $f : \mathcal{M} \longrightarrow \mathbb{R}$ induces a derived map $(df)_p : T_p\mathcal{M} \longrightarrow T_{f(p)}\mathbb{R} \cong \mathbb{R}$ which is a linear function on a vector

$$(df)_p(aX + bY) = a(df)_p(X) + b(df)_p(Y) \quad \text{for } a, b \in \mathfrak{F}(\mathcal{M}) \text{ and } X, Y \in V. \tag{1.46}$$

The function $(df)_p \in V^*$ is a coordinate 1-*form* (a form of degree 1). Given a vector field $X \in V = \mathfrak{X}(\mathcal{M})$, we have

$$(df)_p(X_p) = \frac{df}{dt}(p) = X_p f \in \mathbb{R}. \tag{1.47}$$

Hence, we define a *cotangent bundle* as a set of all cotangent spaces

$$T^*\mathcal{M} = \bigcup_{p\in\mathcal{M}} T_p^*\mathcal{M}. \tag{1.48}$$

The set of all *differential* (or *smooth*) 1-form fields on $\mathcal{M}$ is then denoted by $A^1(\mathcal{M}) = \Gamma(T^*\mathcal{M})$.

Take $f = \varphi^\mu$ and $X_p = (\partial_\nu)_p \in$. The component of local coordinate function is $\varphi^\mu : U \longrightarrow \mathbb{R}$ such that $x^\mu = \varphi^\mu(p)$, then $(d\varphi^\mu)_p : T_pU \longrightarrow T_{x^\mu(p)}\mathbb{R} \cong \mathbb{R}$. Consequently we have property

$$(d\varphi^\mu)_p\big((\partial_\nu)_p\big) = (\partial_\nu)_p(\varphi^\mu)_p = \delta^\mu_\nu. \tag{1.49}$$

The equation above implies that $V^* = T^*\mathcal{M}$ is the dual space of $V = T\mathcal{M}$ and $\{\vartheta^\mu\} = \{d\varphi^\mu\} \equiv \{dx^\mu\}$ forms a coordinate frame in $T^*\mathcal{M}$. Consequently, we can perform an inner product between $T\mathcal{M}$ and $T^*\mathcal{M}$:

$$\langle\ ,\ \rangle : T^*\mathcal{M} \times T\mathcal{M} \longrightarrow \mathbb{R}, \tag{1.50}$$

$$(\omega, X) \longmapsto \langle \omega, X\rangle. \tag{1.51}$$

Let $X = e_j$ and $\omega = \vartheta^i$, we have

$$\boxed{\langle \vartheta^i, e_j\rangle \equiv \vartheta^i(e_j) = \delta^i_j.}\ ^{\dagger 1} \tag{1.52}$$

1.4 Induced maps of Diffeomorphism

Induced maps of vectors and differential 1-forms. Under a *coordinate transformation* between two charts of a point p (1.5), the transformations of natural basis of vectors and differential 1-forms can be understood by using the *chain rule*

$$\begin{cases} \dfrac{\partial}{\partial x_i^\mu} = \dfrac{\partial}{\partial x_j^\nu}\dfrac{\partial \varphi_{ji}^\nu}{\partial x_i^\mu} \equiv \dfrac{\partial}{\partial x_j^\nu}\dfrac{\partial x_j^\nu}{\partial x_i^\mu}, & (1.53\text{a}) \\[2ex] dx_i^\mu = \dfrac{\partial \varphi_{ij}^\mu}{\partial x_j^\nu}\, dx_j^\nu \equiv \dfrac{\partial x_i^\mu}{\partial x_j^\nu}\, dx_j^\nu. & (1.53\text{b}) \end{cases}$$

Given a diffeomorphism between two manifold $f : \mathcal{M} \longrightarrow \mathcal{N}$ such that $p \longmapsto q = f(p)$, a *pushforward* of vectors is induced

$$f_* : T_p\mathcal{M} \longrightarrow T_q\mathcal{N} = T_{f(p)}\mathcal{N}, \tag{1.54a}$$

$$X_p \longmapsto f_*X_p = (f_*X)_q = (f_*X)_{f(p)}. \tag{1.54b}$$

†1 *Quantum mechanically* a vector and a dual vector introduced in a *Hilbert space* $\mathcal{H}$ are represented by a *bra* vector and a *ket* vector by the *Dirac's notation*, respectively:

$$\boxed{\vec{e}_i^{\,*} \quad \longrightarrow \quad \langle i|} \quad \text{and} \quad \boxed{\vec{e}_j \quad \longrightarrow \quad |j\rangle,} \quad \text{which leads to} \quad \boxed{\langle \vec{e}_i^{\,*}, \vec{e}_j\rangle \quad \longrightarrow \quad \langle i|j\rangle = \delta_{ij}.}$$

However, in addition to the transpose operation T, quantum mechanical bra vectors further need the *complex conjugate* operation $*$, i.e., $\langle i| = |i\rangle^\dagger := \big(|i\rangle^{\mathsf{T}}\big)^*$ where $\dagger$ is called the *Hermitian conjugate* operation.

Let us choose coordinate systems in two manifolds (1.10), the pushforward of a basis vector from one to the other is

$$\left.\frac{\partial}{\partial x^\mu}\right|_p \longmapsto f_*\left(\left.\frac{\partial}{\partial x^\mu}\right|_p\right) = \frac{\partial f^\nu_{yx}}{\partial x^\mu}\left.\frac{\partial}{\partial y^\nu}\right|_{q=f(p)} = \frac{\partial f^\nu_{yx}}{\partial x^\mu}\left.\frac{\partial}{\partial y^\nu}\right|_{f(p)} \equiv \frac{\partial y^\nu}{\partial x^\mu}\left.\frac{\partial}{\partial y^\nu}\right|_{f(p)}. \tag{1.55}$$

Similarly, we can define a map with the *backward* direction with respect to the direction of f called a *pullback* of 1-forms induced by the map f

$$f^* : T^*_q\mathcal{N} \longrightarrow T^*_p\mathcal{M} = T^*_{f^{-1}(q)}\mathcal{M}, \tag{1.56a}$$

$$\omega_q \longmapsto f^*\omega_q = (f^*\omega)_p = (f^*\omega)_{f^{-1}(q)}. \tag{1.56b}$$

Therefore the pullback of a coordinate 1-form is

$$dy^\mu|_q \longmapsto f^*(dy^\mu|_q) = \frac{\partial f^\mu_{yx}}{\partial x^\nu}\left. dx^\nu\right|_{p=f^{-1}(q)} = \frac{\partial f^\mu_{yx}}{\partial x^\nu}\left. dx^\nu\right|_{f^{-1}(q)} \equiv \frac{\partial y^\mu}{\partial x^\nu}\left. dx^\nu\right|_{f^{-1}(q)}. \tag{1.57}$$

As discussions above, we compare properties of diffeomorphism, pushforward and pullback together as follows

$$\text{Diffeomorphism } f: \quad \mathcal{M} \text{ (left)} \longrightarrow \mathcal{N} \text{ (right)}, \tag{1.58a}$$

$$\text{Pushforward } f_*: \quad T_p\mathcal{M} \text{ (left)} \longrightarrow T_p\mathcal{N} \text{ (right)}, \tag{1.58b}$$

$$\text{Pullback } f^*: \quad T^*_p\mathcal{N} \text{ (right)} \longrightarrow T^*_p\mathcal{M} \text{ (left)}. \tag{1.58c}$$

It is apparent that the direction of pushforward f_* is the *same* as diffeomorphism f, and the direction of pullback f^*, however, is the *inversed*. That is why we called the objects with subscript the *covariant* fields and with superscript the *contravariant* fields.

1.5 Tensor Space

Tensor fields. Suppose V is a n-dimensional vector space over $\mathfrak{F}$ and its dual space is V^*. We define (r,s)-type tensor space V^r_s (or $V^{(r,s)}$) by the so-called tensor product

$$V^r_s = V^r \otimes V_s = \underbrace{V \otimes V \otimes \cdots \otimes V}_{r \text{ times}} \otimes \underbrace{V^* \otimes V^* \otimes \cdots \otimes V^*}_{s \text{ times}} \equiv \overset{r}{\otimes} V \overset{s}{\otimes} V^*. \tag{1.59}$$

Take e_μ as the basis of V and ϑ^μ as the basis of V^*. Any element in V^r_s is called a tensor field $K \in V^r_s$ which is a map:

$$K : \underbrace{V^* \times \cdots \times V^*}_{r \text{ terms}} \times \underbrace{V \times \cdots \times V}_{s \text{ terms}} \longrightarrow \mathbb{R}. \tag{1.60}$$

So, we can write K by

$$\boxed{K = K^{\mu_1\cdots\mu_r}{}_{\nu_1\cdots\nu_s} e_{\mu_1} \otimes \cdots \otimes e_{\mu_r} \otimes \vartheta^{\nu_1} \otimes \cdots \otimes \vartheta^{\nu_s},} \tag{1.61}$$

where

$$K^{\mu_1\cdots\mu_r}{}_{\nu_1\cdots\nu_s} = K(\vartheta^{\mu_1}, \ldots, \vartheta^{\mu_r}, e_{\nu_1}, \ldots, e_{\nu_s}). \tag{1.62}$$

Obviously, we have $V^0_0 = \mathfrak{F}$, $V^1_0 = V$ and $V^0_1 = V^*$.

Symmetry of tensors. Given any rank-2 tensor, for example a $(0,2)$-type tensor A_{ij}, we can always decompose it into two terms

$$A_{ij} = \frac{1}{2}\big(A_{ij} + A_{ji}\big) + \frac{1}{2}\big(A_{ij} - A_{ji}\big). \tag{1.63}$$

It is apparent that the first term is symmetric in two indices, while the second one is antisymmetric. We always use the following notations to define the symmetrization and antisymmetrization of a tensor, respectively

$$\begin{cases} A_{(ij)} \coloneqq \dfrac{1}{2!}\big(A_{ij} + A_{ji}\big), & \text{(1.64a)} \\ A_{[ij]} \coloneqq \dfrac{1}{2!}\big(A_{ij} - A_{ji}\big). & \text{(1.64b)} \end{cases}$$

For a $(0,3)$-type tensor A_{ijk}, it leads to the notation of

$$\begin{cases} A_{(ijk)} \coloneqq \dfrac{1}{3!}\big(A_{ijk} + A_{jki} + A_{kij} + A_{jik} + A_{kji} + A_{ikj}\big), & \text{(1.65a)} \\ A_{[ijk]} \coloneqq \dfrac{1}{3!}\big(A_{ijk} + A_{jki} + A_{kij} - A_{jik} - A_{kji} - A_{ikj}\big). & \text{(1.65b)} \end{cases}$$

For a $(r,0)$- or $(0,r)$-type tensor of rank-r should be similar to have a coefficient $1/r!$ due to the permutations. In particular if a $(0,3)$-type tensor is emphasized that the first and third indices are antisymmetric, i.e., antisymmetric in i and k, we can write it as

$$A_{[i|j|k]} \coloneqq \frac{1}{2!}\big(A_{ijk} - A_{kji}\big). \tag{1.66}$$

Coordinate transformation. The chain rule of the differential and the derivative would induce a transformation from one coordinate system, $\{x^j\}$, to another one which is denoted by $\{x^{i'}\} \coloneqq \{y^i\}$. So, one coordinate system is regarded as functions of another, i.e., $\{y^i\} = \{y^i(x)\}$, and vise versa.

$$\begin{cases} dy^i = \dfrac{\partial y^i}{\partial x^j}\, dx^j \coloneqq A^{i'}{}_j \cdot dx^j, & \text{(1.67a)} \\ \dfrac{\partial}{\partial y^i} = \dfrac{\partial}{\partial x^j}\dfrac{\partial x^j}{\partial y^i} \coloneqq \dfrac{\partial}{\partial x^j} \cdot (A^{-1})^j{}_{i'}. & \text{(1.67b)} \end{cases}$$

Here we have defined the partial derivative of one coordinate with respect to another as a transformation matrix, which is indeed the *Jacobian matrix*:

$$\mathsf{A} = \left(A^{i'}{}_j\right) \coloneqq \left(\frac{\partial y^i}{\partial x^j}\right) = \begin{pmatrix} \frac{\partial y^1}{\partial x^1} & \frac{\partial y^1}{\partial x^2} & \cdots & \frac{\partial y^1}{\partial x^n} \\ \frac{\partial y^2}{\partial x^1} & \frac{\partial y^2}{\partial x^2} & \cdots & \frac{\partial y^2}{\partial x^n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y^n}{\partial x^1} & \frac{\partial y^n}{\partial x^2} & \cdots & \frac{\partial y^n}{\partial x^n} \end{pmatrix}, \tag{1.68}$$

and its inverse is

$$\mathsf{A}^{-1} = \left((A^{-1})^j{}_{i'}\right) \coloneqq \left(\frac{\partial x^j}{\partial y^i}\right) = \begin{pmatrix} \frac{\partial x^1}{\partial y^1} & \frac{\partial x^1}{\partial y^2} & \cdots & \frac{\partial x^1}{\partial y^n} \\ \frac{\partial x^2}{\partial y^1} & \frac{\partial x^2}{\partial y^2} & \cdots & \frac{\partial x^2}{\partial y^n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x^n}{\partial y^1} & \frac{\partial x^n}{\partial y^2} & \cdots & \frac{\partial x^n}{\partial y^n} \end{pmatrix}. \tag{1.69}$$

Their products would be the unit matrix:

$$(A^{-1})^i{}_{j'}\,A^{j'}{}_k=\delta^i_k \quad\text{and}\quad A^{k'}{}_i\,(A^{-1})^i{}_{j'}=\delta^{k'}_{j'}. \tag{1.70}$$

Given a total differential of a scalar written in both coordinate systems:

$$d\phi=\phi_{,\,i'}\,dx^{i'}=\phi_{,\,j}\,dx^j. \tag{1.71}$$

We identify its convariant component as a covariant vector $\phi_{,\,i'}(y)=V_{i'}(y)$ or $\phi_{,\,j}(x)=V_j(x)$. Using the chain rule for the derivative, we obtain a coordinate transformation of a *covariant* vector

$$\frac{\partial\phi(x)}{\partial x^j}=\frac{\partial\phi}{\partial y^i}\frac{\partial y^i}{\partial x^j}\quad\text{or}\quad\boxed{V_j(x)=V_{i'}(y)\,A^{i'}{}_j.} \tag{1.72}$$

The inner product of two vectors V and W can be realized as a product of a covariant vector and a contravariant vector, which is an invariant quantity whatever the coordinate system is chosen for the vectors, i.e., $g(V,W)$ should be an invariant function. By using this property, we can define the transformation law of a *contravariant* vector

$$V_j(x)W^j(x)=V_{i'}(y)\,A^{i'}{}_j\,W^{i'}(y)\equiv V_{i'}(y)W^{i'}(y)\quad\Longrightarrow\quad\boxed{W^{i'}(y)=A^{i'}{}_jW^j(x).} \tag{1.73}$$

We find that the Jacobian matrix $\mathsf{A}=(A^{i'}{}_j)$ transforms *contravariant* vector components from old coordinate system to the new one; while, in contrast, it transforms a *covariant* vector components from new coordinate system back in the old one. Also, we have the gradient of a scalar written in both coordinate systems

$$\vec{\nabla}\phi=e_{i'}\,\phi^{,\,i'}(y)=e_j\,\phi^{,\,j}(x). \tag{1.74}$$

It is obvious that both (1.71) and (1.74) have the derivative of $\phi(x)$ with respect to x^j in different representations, which is called *duality*. Due to the duality of $\phi^{,\,j}$ and $\phi_{,\,j}$, we can identify ϑ^j, the dual vector of e_j, as the differential of the coordinate x^j:

$$\left.\begin{array}{ccc}\phi^{,\,j} & \longrightarrow & \vec{\nabla}\phi\\ \phi_{,\,j} & \longrightarrow & d\phi\end{array}\right\}\quad\Longrightarrow\quad e_j\xleftrightarrow{\text{dual basis}}\boxed{\vartheta^j\equiv dx^j,} \tag{1.75}$$

and its transformation satisfies the rule of (1.67a) or (1.73). Generally, we can perform the transformation rule of a (r,s)-type tensor like a tensor constructed by the tensor product

$$\begin{aligned}&\underbrace{\vec{V}\otimes\vec{V}\otimes\cdots\vec{V}}_{r\text{-times}}\otimes\underbrace{\vec{W}^*\otimes\vec{W}^*\otimes\cdots\vec{W}^*}_{s\text{-times}}\\&=V^{i_1}V^{i_2}\cdots V^{i_r}W_{j_1}W_{j_2}\cdots W_{j_s}\,e_{i_1}\otimes e_{i_2}\otimes\cdots\otimes e_{i_r}\otimes dx^{j_1}\otimes dx^{j_2}\otimes\cdots\otimes dx^{j_s}.\end{aligned} \tag{1.76}$$

Identifying $K^{i'_1i'_2\cdots i'_r}{}_{j'_1j'_2\cdots j'_s}$ as the composite tensor $V^{i_1}V^{i_2}\cdots V^{i_r}W_{j_1}W_{j_2}\cdots W_{j_s}$, the coordinate transformation rule of a tensor $K^{i'_1i'_2\cdots i'_r}{}_{j'_1j'_2\cdots j'_s}$ should be

$$\boxed{K^{i'_1i'_2\cdots i'_r}{}_{j'_1j'_2\cdots j'_s}=A^{i'_1}{}_{i_1}A^{i'_2}{}_{i_2}\cdots A^{i'_r}{}_{i_r}K^{i_1i_2\cdots i_r}{}_{j_1j_2\cdots j_s}(A^{-1})^{j_1}{}_{j'_1}(A^{-1})^{j_2}{}_{j'_2}\cdots(A^{-1})^{j_s}{}_{j'_s},} \tag{1.77}$$

which is a *homogeneous* transformation.

Covariance (the transformation law of tensor fields)

It is important to note that a tensor can be expressed both in two coordinate systems by

$$\begin{aligned} K &= K^{i'_1 i'_2 \cdots i'_r}{}_{j'_1 j'_2 \cdots j'_s}\, e_{i'_1} \otimes e_{i'_2} \otimes \cdots \otimes e_{i'_r} \otimes dy^{j'_1} \otimes dy^{j'_2} \otimes \cdots \otimes dy^{j'_s} \\ &= K^{i_1 i_2 \cdots i_r}{}_{j_1 j_2 \cdots j_s}\, e_{i_1} \otimes e_{i_2} \otimes \cdots \otimes e_{i_r} \otimes dx^{j_1} \otimes dx^{j_2} \otimes \cdots \otimes dx^{j_s}, \end{aligned} \tag{1.78}$$

where two components are related by

$$K^{i'_1 i'_2 \cdots i'_r}{}_{j'_1 j'_2 \cdots j'_s} = A^{i'_1}{}_{i_1} A^{i'_2}{}_{i_2} \cdots A^{i'_r}{}_{i_r} K^{i_1 i_2 \cdots i_r}{}_{j_1 j_2 \cdots j_s} (A^{-1})^{j_1}{}_{j'_1} (A^{-1})^{j_2}{}_{j'_2} \cdots (A^{-1})^{j_s}{}_{j'_s}. \tag{1.79}$$

It is similar to the inner product of two vectors that all the indices of components are contracted with the indices of basis and dual basis vectors. Therefore, the form of any tensor fields should be invariant in any coordinate systems. Physically, we would express physical laws in the form of tensor equations, which implies that physical laws should be the *same* in all coordinate systems. Moreover, the transformation law of tensor fields reflects that laws of physics are *covariant* under the coordinate transformation.

1.6 1-Parameter Group of Transformations

A 1-parameter group. Consider a map given by

$$\sigma : \mathbb{R} \times \mathcal{M} \longrightarrow \mathcal{M}, \tag{1.80}$$

$$(t, p) \longmapsto \sigma(t, p). \tag{1.81}$$

We can construct an *automorphism* of $\mathcal{M}$ by $\varphi_t(p) \coloneqq \sigma(t, p)$ for fixed t

$$\varphi_t : p \in \mathcal{M} \longmapsto q = \varphi_t(p) \in \mathcal{M}. \tag{1.82}$$

For any $t \in \mathbb{R}$, the map φ_t is called the 1-parameter group of (differentiable) transformations of $\mathcal{M}$, which satisfies the following conditions:

- $\varphi_0 = \mathrm{id}_{\mathcal{M}}$, i.e., $\varphi_0(p) = \sigma(0, p) = p$;
- $\varphi_t \circ \varphi_s = \varphi_{t+s}$ for any $t, s \in \mathbb{R}$.

The second condition can be realized as follows:

$$(\varphi_t \circ \varphi_s)(p) = \gamma_{\varphi_s(p)}(t) = \gamma_p(t + s) = \sigma(t + s, p) = \varphi_{t+s}(p). \tag{1.83}$$

By substituting $s = -t$ in the second condition, it is easy to show that $\varphi_{-t} = \varphi_t^{-1}$. The $\varphi_t : \mathcal{M} \longrightarrow \mathcal{M}$ is a *diffeomorphism* from $\mathcal{M}$ to itself, which forms a diffeomorphism group denoted by $\mathrm{Diff}(\mathcal{M})$. This diffeomorphism also induces a map of tangent vectors $\varphi_{t*} : T\mathcal{M} \longrightarrow T\mathcal{M}$:

$$\varphi_{t*}(X_p) = X_q = X_{\varphi_t(p)}. \tag{1.84}$$

It is obvious that $\gamma_p(t) \coloneqq \sigma(t, p)$ for fixed p can be regarded as a curve passing throug the point p:

$$\gamma_p : t \in \mathbb{R} \longmapsto \gamma_p(t) \in \mathcal{M}. \tag{1.85}$$

Let X be a vector field on $\mathcal{M}$, a parameterized curve γ_p called the *orbit* of φ_t through $p \in \mathcal{M}$. The curve $\varphi_t(p)$ is called an *integral curve* of X through the point p at $t = 0$, and the map φ_t sends one point to another is called the *flow*) generated by X. If a vector field X is induced by the transformation, we call it the *complete* vector field on $\mathcal{M}$. Conversely, any $X \in \mathfrak{X}(\mathcal{M})$ can generate a 1-parameter group of transformations of $\mathcal{M}$, $\varphi_t \in \mathrm{Diff}(\mathcal{M}) : \mathcal{M} \longrightarrow \mathcal{M}$.

A *local* version of transformation can be defined in a same way, if t is defined in a neighborhood of 0 and p is in an open set U of $\mathcal{M}$. The induced vector field is

$$X(t,p) \equiv X_{\gamma_p(t)} = d\gamma_p\left(\frac{d}{dt}\right) = \frac{d\gamma_p(t)}{dt} = \frac{d}{dt}\sigma(t,p). \tag{1.86}$$

Let x be a local coordinates of U, then the vector component in U should be

$$X^\mu(t,p) = X_{\gamma_p(t)}(x^\mu) = d\gamma_p\left(\frac{d}{dt}\right)(x^\mu) = \frac{d}{dt}\big(x^\mu \circ \gamma_p(t)\big) = \frac{d}{dt}x^\mu\big(\sigma(t,p)\big). \tag{1.87}$$

Similarly, any $X \in \mathfrak{X}(U)$ generates a local 1-parameter group of *local* transformations of U: $\varphi_t \in \mathrm{Diff}(U) : U \longrightarrow U$. By Taylor expansion, we have a local diffeomorphism from a point p to $\sigma(t,p) \equiv q$:

$$q = \sigma(t,p) = \sigma(0,p) + t\,X(0,p) + \mathcal{O}(t^2) = p + t\,X_p + \mathcal{O}(t^2), \tag{1.88}$$

which can be expressed in the local coordinate system in linear term

$$x^\mu(t,p) = x^\mu(0,p) + t\,X^\mu(0,p) \quad \text{or} \quad x^\mu(q) = x^\mu(p) + t\,X^\mu_p, \tag{1.89}$$

where we have defined $x^\mu(t,p) \coloneqq x^\mu\big(\sigma(t,p)\big) \equiv x^\mu(q)$.

2 Exterior Differential Calculus

In preparation.